TAMBALI SATHVIKA

Computer Science & Data Science Undergraduate • Full-Stack Developer

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PROFESSIONAL SUMMARY

Passionate and solution-oriented Computer Science & Data Science undergraduate at Mohan Babu University with hands-on expertise in Data Structures, Algorithmic Problem Solving, Full-Stack Web Development, and Automated Workflow Engineering. Proficient in building interactive analytics platforms with React and Python, designing end-to-end webhook pipelines via n8n, and conducting statistical data wrangling with Pandas and Matplotlib. Demonstrated aptitude for clean architectural design, continuous learning, and scalable software delivery.

EDUCATION

Mohan Babu University (MBU) — **Tirupati, Andhra Pradesh**
B.Tech in Computer Science & Engineering (Data Science) • Roll No: 24102A030126 — 2024 – 2028
• Continuous Evaluation Grade: **A+ (Outstanding, 29/30 CIE)** • Coursework: Data Structures, Python for Data Science, Discrete Mathematics, Relational DBMS.

Junior College (Board of Intermediate Education, AP) — **Andhra Pradesh**
Higher Secondary Certificate — Mathematics, Physics, Chemistry (MPC) — 2022 – 2024

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Data Science & ML:** Pandas, NumPy, Matplotlib, Seaborn, SQLite, Data Ingestion, EDA, Missing Data Imputation, Outlier Detection (IQR), Time Series
- **Web Technologies:** React, Next.js, Tailwind CSS, RESTful APIs, JSON, Responsive UI/UX, Dynamic Routing
- **Automation & Tools:** n8n Workflow Automation, Google Sheets API, Gmail Webhooks, Git, GitHub, VS Code, Postman, Vercel
- **Core Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Normalization, Time/Space Complexity

KEY PROJECTS

Data Science Digital Laboratory & Interactive Kernel [Live Demo] [GitHub]
Tech Stack: React, Vite, Python, JavaScript, Tailwind CSS, Vercel
- Architected and deployed an interactive academic web platform organizing 6 university laboratory modules and 23 verified experimental tasks.
- Engineered an embedded multi-language code editor supporting 12 programming languages with live syntax diagnostics and execution buffers.
- Integrated interactive visual plot previews, dynamic Viva-Voce technical interview modules, and responsive external video lectures.

Service Request & Lead Automation Pipeline [GitHub]
Tech Stack: n8n Workflow Automation, Webhooks, Google Sheets API, Gmail API, JSON
- Constructed an automated backend integration pipeline capturing webhook triggers from client inquiry forms with sub-300ms response time.
- Structured data validation and automated CRM record appending into Google Sheets, coupled with dynamic personalized Gmail alert dispatches.
- Designed error handling, payload normalization, and automatic retry policies achieving zero lead loss across automated operations.

Personal Developer Portfolio & Digital Showcase [Live Demo] [GitHub]
Tech Stack: Next.js, React, TypeScript, Tailwind CSS, Lucide Icons, Vercel
- Created a responsive developer portfolio showcasing featured web projects, technical competencies, verified credentials, and interactive contact channels.
- Achieved 95+ Google Lighthouse performance, accessibility, and SEO scores through server-side optimization and component modularization.

Artisan Cafe & Bistro Web Application [Live Demo] [GitHub]
Tech Stack: HTML5, CSS3, JavaScript (ES6+), Mobile-First UI/UX
- Developed a modern, responsive small-business web portal featuring categorized food & beverage menus, customer review carousel, and table reservation booking flow.

Data Structures & Algorithms Repository [GitHub]
Tech Stack: C++, Python, DSA, Competitive Programming
- Implemented 70+ optimal algorithmic solutions across arrays, linked lists, binary trees, recursion, and dynamic programming with full space-time complexity analysis.

CERTIFICATIONS, LEADERSHIP & HONORS

- **Continuous Internal Evaluation (CIE) Distinction:** Scored 29/30, securing the highest performance band in the Department of Data Science at MBU.
- **Active Competitive Programmer:** Regular problem solver on LeetCode with continuous practice in arrays, hashing, and pointer manipulation.
- **Open Source Contributor:** Maintained 10+ public GitHub repositories spanning Java, Python practicals, full-stack web apps, and automation workflows.